

Name: Srinidhi A
ID: COMETFWC053

Question 59

Two products are sold from a vending machine, which has two push buttons P_1 and P_2 . When a button is pressed, the price of the corresponding product is displayed in a 7-segment display.

- If no buttons are pressed, '0' is displayed, signifying Rs. 0.
- If only P_1 is pressed, '2' is displayed, signifying Rs. 2.
- If only P_2 is pressed, '5' is displayed, signifying Rs. 5.
- If both P_1 and P_2 are pressed, 'E' is displayed, signifying 'Error'.

Consider:

- (a) push button pressed/not pressed $\rightarrow$ logic 1/0.
- (b) segment glowing/not glowing $\rightarrow$ logic 1/0.

If a segment output is considered as a function of P_1 and P_2 , which of the following is correct?

1. $g = \overline{P_1} + P_2, \quad d = c + e$
2. $g = P_1 + P_2, \quad d = c + e$
3. $g = \overline{P_1} + P_2, \quad e = b + c$
4. $g = P_1 + P_2, \quad e = b + c$

Solution

From the display requirements:

P_1	P_2	Output
0	0	0
1	0	2
0	1	5
1	1	E

From the standard 7-segment patterns:

Character	a	b	c	d	e	f	g
0	1	1	1	1	1	1	0
2	1	1	0	1	1	0	1
5	1	0	1	1	0	1	1
E	1	0	0	1	1	1	1

Observing segment **g**:

It is OFF only when $P_1 = 0$ and $P_2 = 0$. Hence,

$$g = P_1 + P_2$$

Observing segment **e**:

It is ON in cases corresponding to segments b or c . Hence,

$$e = b + c$$

Correct Option: 4